

**AI can solve math word problems**

A system that solves grade school math problems with nearly twice the accuracy of a GPT-3 model. <https://digit.in/nov21-31>

**Intel's new neuron-like processors**

Intel launches its next-gen processor that has electronics that behaves a lot like neurons. <https://digit.in/nov21-32>

Biotech Startups

The other revolution

Aditya Madanapalle | aditya@digit.in

T

he Biotechnology Industry Research Assistance Council's (BIRAC) Sustainable Entrepreneurship

and Enterprise Development Fund (SEED) funding program have awarded a total of 75 lakh to three startups, Beaniesnak, Savemom and Inphlox Water Systems. These three are respectively involved in making high tech nachos, tackling the challenging problem of neonatal healthcare, and developing an innovative decentralised as well as modular wastewater treatment system. We spoke to these startups to understand more about the problems that they are trying to solve, and how they are going about it.

NEONATAL CARE

The healthcare of new mothers and their just born babies is an area severely lacking in the country, and Savemom is doing its bit to tackle this difficult problem with technology. The first question we asked was about the Allowear wearable, and how it works. "A smart wearable device that continuously tracks activity, sleep cycle reminds to take medicine, checkup and that increases the interaction of the patient. The device has more than six-month battery life. The Nail form factor PCBA design allows the user to change the wearable design based on



The Allowear

their needs. Inbuilt memory allows storing the user data throughout pregnancy time. On the Device, a security chip enables to the production of the data & acts as an authentication device to access the patient medical information." We then asked about the AlloTricorder, and its functions. "An IoT-based kit that measures 6 vital pieces of information in a single device that sends the data over to the cloud for analytics. The device integrates a blood oxygen sensor, infrared body temperature sensor, single lead ECG test module into a compact monitor, which helps to measure the above parameters and enables users to enjoy basic medical service within 10 minutes irrespective of place. AlloTricorder establishes a Bluetooth connection with the application and uses cloud storage to maintain and transfer the data." Apart from the wearables, there are also an app that is used by social workers, doctors and even end users. We asked how this app

works, "the Allowear wearable device will be given to the pregnant women for continuous monitoring of her activities and give timely reminders for medicine. The AlloTricorder an IoT based multi vital monitoring device along with the smartphone will be given to the health worker. The husband's mobile number will be configured for the notification and follow up. Every month the health worker will visit the pregnant women home with their medical device and smartphone. The phone will identify the pregnant mother by connecting the Allowear wearable to the device. It syncs the stored activity data from the device to a mobile phone through Bluetooth. Using the AlloTricorder medical device the health worker digitally collects 10 vital parameters including (Blood Pressure, heart rate, SpO2, temperature, ECG, Heart rate variability) of pregnant women and sends the data over to the cloud for doctor feedback. Through the